

Course Code: ES590	Course Name: Advanced Electrodynamics	Credit Hours: 3
Course Instructor: Sheharyar Pervez (G-27, FBS) Email: sheharyar@giki.edu.pk		Graduate Course
Aims and Objectives		
Maxwell's equations are considered the most beautiful in physics because they unite electricity, magnetism, and light into a single, elegant framework using just four short vector calculus statements. This course (core requirement for the MS in Engineering Science, Applied Physics program) provides a rigorous and advanced treatment of classical electrodynamics, developing the conceptual and mathematical foundations needed to understand how electric and magnetic fields shape the physical world. Beginning with a unified review of static fields and boundary-value techniques, the curriculum progresses naturally to dynamic fields, relativistic mechanics, and the fully covariant formulation of electrodynamics. Along the way, students develop a deeper appreciation of the unity, symmetry, and elegance underlying electromagnetic phenomena.		
Course Contents		
Review of Static Fields Boundary-Value Problems & Special Functions Maxwell's Equations & Conservation Laws Electromagnetic Waves & Radiation Relativistic Electrodynamics & Covariant Formulation		
Course Learning Outcomes		
CLO 1: Formulate classical electrodynamics in four-vector notation to analyze electromagnetic fields, potentials, and conservation laws within a fully Lorentz-covariant relativistic framework.		
CLO 2: Apply mathematical techniques to solve boundary-value problems in electrostatics and magnetostatics across various coordinate systems.		
Week	Topic	
Week 1-2	Mathematical and physical review	
Week 3-5	Relativistic concepts and the covariant formulation of electrodynamics	
Week 7-10	Boundary-value problems in electrostatics	
Week 11-13	Boundary-value problems in magnetostatics	
Week 14-15	EM Radiation	
Recommended Textbooks		
Text Books		
1. <i>Classical Theory of Fields</i> , Landau & Lifshitz.		
2. <i>Electrodynamics: An Intensive Course</i> . Chaichian et al. Springer (2016)		
3. <i>Classical Electrodynamics</i> by J. D. Jackson, 3 rd Ed. (John Wiley & Sons, N.Y).		
Reference Books		
4. <i>Electricity and Magnetism</i> , Edward M. Purcell.		
5. <i>Introduction to Electrodynamics</i> , David J. Griffiths, the usual textbook for a first course in electrodynamics		
6. <i>Classical Electromagnetic Radiation</i> , Mark A. Heald and Jerry B. Marion		
7. <i>A history of the theories of aether and electricity from the age of Descartes to the close of the 19th century</i> , E.T. Whittaker, selected specifically for the curious.		